Omega-consistent theories do not refute their Godel sentence

The Open Logic Project contributors

openlogic:incompleteness/incompleteness-provability:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author lemma labelled lem:omega-cons-G-unref (517-520); the proof span 522-541 is the author proof environment that immediately follows it.

Editorial note. External reference numbers render the unit-internal labels with hyphens in place of colons (pipeline number charset); the Source reference lines below the body give each label's exact source span.

Editorial scope. One original result of the Open Logic Project unit incompleteness/incompleteness-provability, counted once: omega-consistent theories do not refute their Godel sentence. Only the counted lemma and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Omega-consistency. A theory is omega-inconsistent when for some formula A(x) it proves the existential 'there is x with A(x)' while also proving each negated instance A(n-bar); omega-consistent theories are consistent.. The proof concludes by displaying exactly this pattern for the proof predicate (536-541).
- The Godel sentence and its Q-form. The negated Godel sentence is equivalent in Q to the existential 'there is x with Prf_T(x, gG_T)' (the unit's equation eqn:qpf).. It converts the assumed derivation of the negated Godel sentence into a derivation of the existential (538-540).
- Consistent theories do not prove the Godel sentence. If T is consistent then T does not derive G_T; the unit proves this as its own lemma.. It supplies, in the consistent case, the non-derivability from which the numeral instances are refuted (528-531).
- Representability of the proof predicate in Q. For each n, the fact that there is no derivation of a given sentence coded n is witnessed in Q by a derivation of the negated proof predicate instance Prf_T(n-bar, gG_T); this representability standard of the unit underwrites the infinite family.. It yields the displayed family of Q-derivations, hence T-derivations, of every negated numeral instance (531-537).

Source reference. Original lemma lem-cons-G-unprov: sources/olp-incompleteness-incompleteness-provability/pinned-original-source.tex, source lines 484–484. The author proof cites the unit's lemma that consistent theories do not prove their Godel sentence. The load-bearing work is carried by that separately proved lemma of the same unit; the counted proof retains its own omega-inconsistency construction in full.

Source reference. Original equation eqn-qpf: sources/olp-incompleteness-incompleteness-provability/pinned-original-source.tex, source lines 478–478. The author proof cites the unit's equation giving the Q-equivalent existential form of the negated Godel sentence. It is a navigation pointer to the displayed equivalence of the same unit; the counted proof retains its own argument in full.

Lemma 0.1. *If* ThT *is an* ω*-consistent, !!axiomatizable theory extending* ThQ*, then* Th$T \vdash /\neg!G_{\mathrm{Th}}T$.

Proof. We show that if ThT !!derives $\neg!G_{\mathrm{Th}}T$, then it is ω-inconsistent. Suppose ThT !!derives $\neg!G_{\mathrm{Th}}T$. If ThT is inconsistent, it is ω-inconsistent, and we are done. Otherwise, ThT is consistent, so it does not !!derive $!G_{\mathrm{Th}}T$ by lem-cons-G-unprov. Since there is no !!derivation of $!G_{\mathrm{Th}}T$ in ThT, ThQ !!derives

$$\neg\,\mathrm{Prf}[\mathrm{Th}T](\underline{0}, \mathrm{gn}\,!G_{\mathrm{Th}}T), \neg\,\mathrm{Prf}[\mathrm{Th}T](\underline{1}, \mathrm{gn}\,!G_{\mathrm{Th}}T), \neg\,\mathrm{Prf}[\mathrm{Th}T](\underline{2}, \mathrm{gn}\,!G_{\mathrm{Th}}T), \ldots$$

and so does ThT. On the other hand, by eqn-qpf, $\neg!G_{\mathrm{Th}}T$ is equivalent to $\exists[x][\mathrm{Prf}[\mathrm{Th}T](x, \mathrm{gn}\,!G_{\mathrm{Th}}T)]$. So Th$T$ is ω-inconsistent. □

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$$\neg\,\mathrm{Prf}[\mathrm{Th}T](\underline{0}, \mathrm{gn}\,!G_{\mathrm{Th}}T), \neg\,\mathrm{Prf}[\mathrm{Th}T](\underline{1}, \mathrm{gn}\,!G_{\mathrm{Th}}T), \neg\,\mathrm{Prf}[\mathrm{Th}T](\underline{2}, \mathrm{gn}\,!G_{\mathrm{Th}}T), \ldots$$

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